

SECURITY OPERATIONS CENTER (SOC) ANALYSIS REPORT

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Target: SOC Sample Logs Analysis — File: SOC_Task2_Sample_Logs

Tools Used: Splunk (SPL Queries, Dashboards)

Track Code: FUTURE_CS_02

1. Executive Summary

This report documents the analysis of **SOC_Task2_Sample_Logs** as part of Task 2. Using Splunk queries and manual review, a total of **50 events** were parsed, categorized, and analyzed.

Key detections include:

- **Ransomware behavior**
- **Rootkit detections**
- **Multiple Trojan infections**
- **Suspicious login sequence** indicating possible credential compromise

These findings highlight **multiple high-priority incidents** that require immediate containment, investigation, and remediation.

2. Scope & Objectives

Scope: Analysis of SOC_Task2_Sample_Logs containing system, authentication, and malware detection logs.

Objective: Extract, analyze, and prioritize suspicious activities, and provide recommended response actions.

3. Environment & Setup

- **Platform Used:** Splunk SIEM
- **Ingested Data:** SOC_Task2_Sample_Logs
- **Queries Written:** For event counts, malware detection, suspicious login analysis, and IP activity
- **Dashboards:** Created for event visualization, top IPs, and login sequences

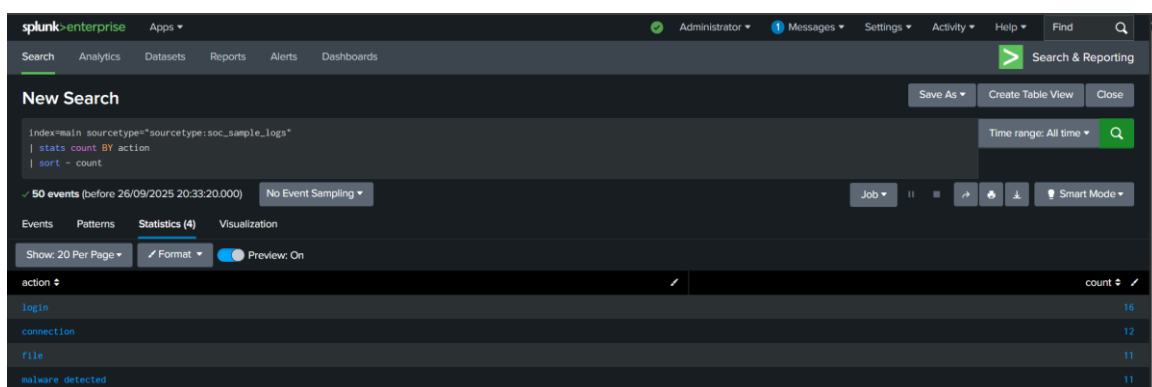
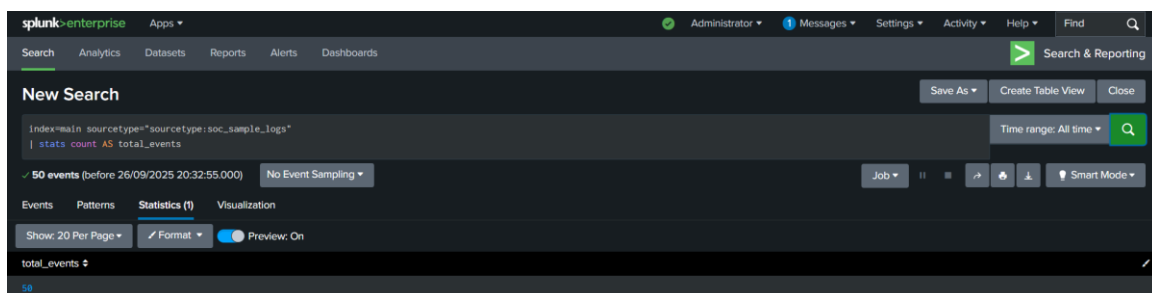
4. Analysis Methodology

1. Logs ingested into **Splunk index=main** with sourcetype set to soc_sample_logs.
2. **SPL queries** executed for event categorization and malware distribution.
3. **Regex queries** used to identify failed → success login sequences.
4. Correlated **suspicious IP activity** with malicious actions.
5. Reviewed **timeline of events** to identify potential attack scenarios.

5. Findings (Detailed)

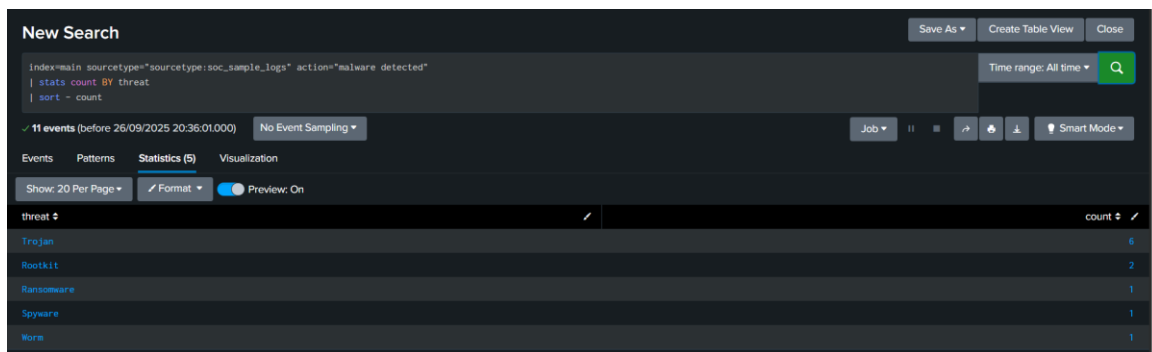
A. Event Overview

- **Total Events: 50**
- **Breakdown:**
 - Connection attempts: 12
 - Login success: 11
 - File accessed: 11
 - Malware detected: 11
 - Login failed: 5



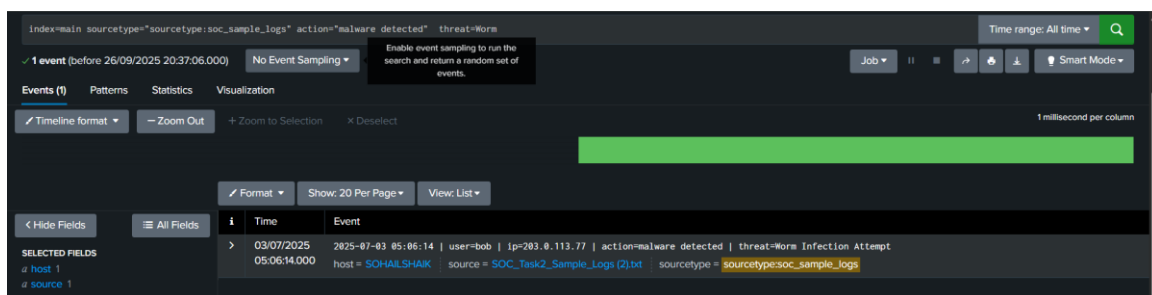
B. Malware Events

- **Total Malware Alerts: 11**
- **Threat Distribution:**
 - Trojan Detected: 6
 - Rootkit Signature: 2
 - Spyware Alert: 1
 - Worm Infection Attempt: 1
 - Ransomware Behavior: 1
- **Top Affected Users: bob (3), alice (3), eve (3)**



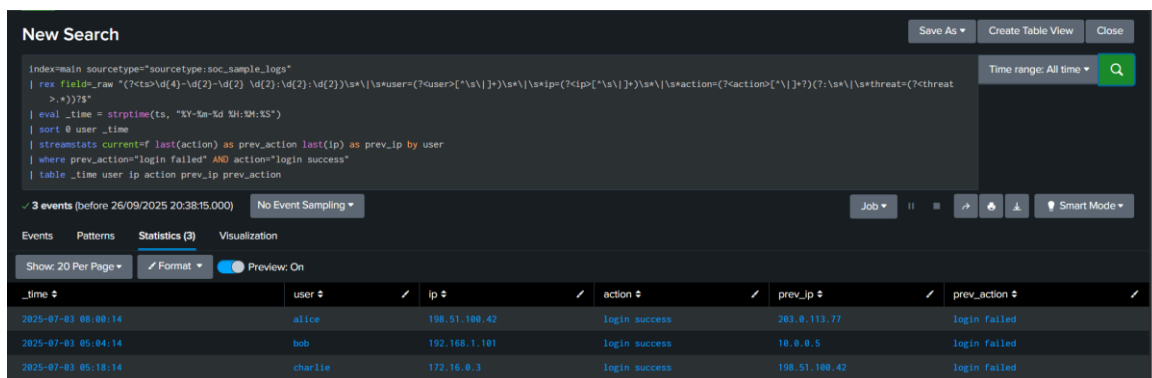
C. Suspicious IP Activity

- IP **203.0.113.77** recorded **15 events** — much higher than normal, suggesting probing/malicious behavior.



D. Failed → Success Login Sequences

- User **bob** had:
 - Failed login at *2025-07-03 04:47:14* from 10.0.0.5
 - Successful login at *2025-07-03 05:04:14* from 192.168.1.101 (≈17 min later)
- This is **suspicious** and may indicate credential compromise.



6. Suspicious Alerts & Prioritization

- Ransomware Behavior (bob, 2025-07-03 09:10:14)**
 - Priority: **High**
 - Action: Isolate host, collect disk image, run EDR, block suspect IPs
- Rootkit Signature (alice, eve)**

- Priority: **High**
- Action: Quarantine hosts, collect memory images, escalate to IR team
- **Multiple Trojans (bob, alice, eve)**
 - Priority: **Medium-High**
 - Action: Identify common file/hash, scan endpoints, check lateral movement
- **Worm Infection Attempt (bob, from 203.0.113.77)**
 - Priority: **Medium**
 - Action: Monitor for scanning, block IP if confirmed malicious
- **Failed → Success Login Sequence (bob)**
 - Priority: **Medium-High**
 - Action: Force password reset, review sessions, investigate activity

Alert	Priority
Ransomware Behavior	High
Rootkit Signature	High
Trojan Detected	Medium-High
Worm Infection Attempt	Medium
Failed→Success Login (bob)	Medium-High

7. Risk Assessment & Recommended Actions

Priority 1 (Immediate):

- Contain ransomware and rootkit-affected hosts
- Quarantine infected systems, preserve forensic evidence

Priority 2 (Short Term):

- Reset compromised accounts, enforce MFA
- Block malicious IPs, strengthen network monitoring

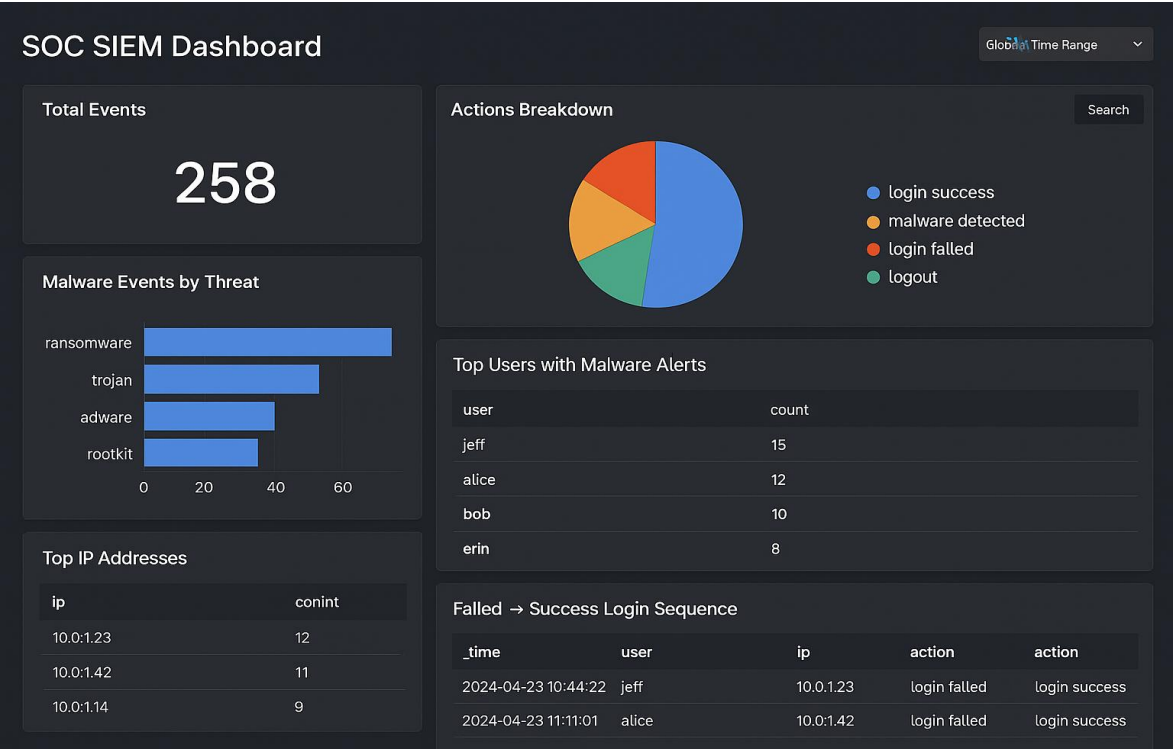
Priority 3 (Medium Term):

- Conduct organization-wide malware scans
- Enhance SIEM rules, improve alert thresholds
- Provide user security awareness training

8. Conclusion

The SOC analysis of Task 2 revealed **multiple high-severity alerts** including ransomware, rootkits, and trojans affecting critical users (bob, alice, eve). One IP (**203.0.113.77**) showed abnormal activity. A suspicious login anomaly indicates potential **credential compromise**.

Immediate containment and escalation are required, alongside long-term **security hardening** measures.



9. Appendix — SPL Commands

index=main sourcetype="soc_sample_logs" | stats count AS total_events

index=main sourcetype="soc_sample_logs" | stats count BY action

index=main sourcetype="soc_sample_logs" action="malware detected" | stats count BY threat

index=main sourcetype="soc_sample_logs" action="malware detected" | stats count BY user

index=main sourcetype="soc_sample_logs" | stats count BY ip

index=main sourcetype="soc_sample_logs" ... (regex query for failed → success login sequence)